

Witness-independence map (v427): disjoint grammars multiply, one-generator projections compress

Gauss $\mathbb{Z}[i]$

$$N(3+2i) = 13$$

Eisenstein $\mathbb{Z}[\omega]$

$$N(3+2\omega) = 7$$

Cyclotomy $\mathbb{Q}(\zeta_5)$

$$N(3+2\zeta_5) = 55$$

Galois $(\mathbb{Z}/5)^\times$

$$|\cdot| = 4$$

Root lattice E_8

$$|\det \text{Cartan}| = 1$$

Pascal/exterior

$$C(4, \leq 2) = 11, 2^4 = 16$$

Coxeter

$$\varphi(30) = 8$$

carrier
 $\{c_3, g_{\text{car}}\}$

disjoint grammars
 $\Rightarrow$ **MULTIPLY**

anchor $a = (1, 1, 2)$
 $\rightarrow (4, 5, 2), (240, 248, 8)$:
ONE generator $\Rightarrow$ **COMPRESS**